

Generation Interconnection System Impact Study Report

For Queue Project Q14-0312

Vonore 345 kV
October 18, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0312. The Interconnection Customer proposes a 1129.2 MW natural gas combined cycle facility interconnecting at the Vonore 345 kV substation in the West Virginia area. The request was received on February 28, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 173 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$134,024,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Vasona Substation – Excell 138 kV L	Morgantown Energy Facility – Vonore 345 kV	548.5	198	100.0
Oakely – Vasona 138 kV	Morgantown Energy Facility – Vonoe 345 kV	538.1	202	100.0
Oakely – W Murray 138 kV	Vonore – Morgantown Energy Facility 345 kV	529.8	206	100.0
Titans – W Murray 138 kV	Vonore – Morgantown Energy Facility 345 kV L	524.8	208	100.0
Titans – Vonore 138 kV	Vonore Substation – Morgantown Energy Facility 345 kV Line	484.4	229	100.0
Excell – Coalinga 1 138 kV line	Vonore – Morgantown Energy Facility 345 kV	396.5	181	62.9
Vonore 138 kV/1 kV Transformer Bank	Morgantown Energy Facility – Vonore 345 kV	343.3	323	100.0
Vonore 345/1 kV Autotransformer	Morgantown Energy Facility – Vonore 345 kV	317.9	349	100.0
Carpenter – Excell Substation 138 kV	Vonore – Morgantown Energy Facility 345 kV	205.0	175	37.1
Carpenter – Landis Sta 138 kV	Vonore – Morgantown Energy Facility 345 kV	197.0	181	37.1
Market – Boulder Crk 138 kV L	Vonore – Morgantown Energy Facility 345 kV L	187.0	202	32.6
Dresden Drive – Landis Sta 138 kV	Morgantown Energy Facility – Vonore 345 kV	184.4	191	37.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Coalinga 1 – Boulder Creek 138 kV line	Vonore – Morgantown Energy Facility 345 kV	182.6	210	32.6
Delta Person Llc – Dresden Drive 138 kV line	Vonore – Morgantown Energy Facility 345 kV	173.6	200	37.1
Pit 7 Tap – Fort Martin 138 kV L	Morgantown Energy Facility – Prides Corner 345 kV	171.6	274	20.6
Fort Martin – Longview Power Plant 138 kV	Morgantown Energy Facility – Prides Corner 345 kV	167.4	288	20.6
Coalinga 1 – Fitzwilliam 138 kV	Vonore – Morgantown Energy Facility Substation 345 kV Line	156.5	211	30.3
Auburn Rd – Prides Corner 345 kV	Firstenergy Ft Martin Power Station – Ronco 345 kV	138.0	1,244	55.2
Westward – Auburn Rd 138 kV L	Auburn Road – Prides Corner 345 kV L	130.6	201	6.3
Tuco Substation – John Day Substation 138 kV	Brwell – Youngwood 500 kV	128.4	231	5.9
Matanzas Substation – John Day Substation 138 kV	Burwell – Youngwood 500 kV	126.8	240	5.9
Auburn Road 500/345 kV TX	Acec Dellwood – Auburn Rd 345 kV	126.7	676	25.4
Prides Corner – Morgantown Energy Facility 345 kV line	Greenland Gap – Santee Coper 345 kV	124.4	1,168	66.3
Ronco 500 kV/345 kV Autotransformer	Auburn Rd – Prides Corner 345 kV	120.6	613	19.5
Westward – Cavendish 138 kV	Prides Corner – Auburn Rd 345 kV	118.8	226	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vasona Substation – Excell 138 kV L	Rebuild line with higher-capacity conductor	\$3,255,000	100% (\$3,255,000)
Oakely – Vasona 138 kV	Rebuild line with higher-capacity conductor	\$2,603,000	100% (\$2,603,000)
W Murray – Oakely 138 kV	Rebuild line with higher-capacity conductor	\$6,161,000	100% (\$6,161,000)
W Murray – Tians 138 kV	Rebuild line with higher-capacity conductor	\$4,466,000	100% (\$4,466,000)
Titans – Vonore 138 kV	Rebuild line with higher-capacity conductor	\$6,070,000	100% (\$6,070,000)
Excell – Coalinga 1 138 kV Line	Rebuild line with higher-capacity conductor	\$5,483,000	99% (\$5,418,000)
Vonore 138/1kV Transformer Bank	Replace transformer with higher-rated unit	\$4,229,000	100% (\$4,229,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vonore 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$20,684,000	100% (\$20,684,000)
Carpenter – Excell Substation 138 kV	Rebuild line with higher-capacity conductor	\$4,024,000	98% (\$3,943,000)
Carpenter – Landis Sta 138 kV	Rebuild line with higher-capacity conductor	\$1,915,000	98% (\$1,877,000)
Market – Boulder Crk 138 kV L	Rebuild line with higher-capacity conductor	\$5,145,000	100% (\$5,145,000)
Dresden Drive – Landis Sta 138 kV	Rebuild line with higher-capacity conductor	\$5,175,000	98% (\$5,071,000)
Coalinga 1 – Boulder Creek 138 kV L	Rebuild line with higher-capacity conductor	\$2,558,000	100% (\$2,558,000)
Delta Person Ic – Dresden Drive 138 kV L	Rebuild line with higher-capacity conductor	\$6,978,000	98% (\$6,838,000)
Pit 7 Tap – Fort Martin 138 kV	Rebuild line with higher-capacity conductor	\$3,358,000	56% (\$1,891,000)
Fort Martin – Longview Power Plant 138 kV	Rebuild line with higher-capacity conductor	\$892,000	56% (\$502,000)
Coalinga 1 Substation – Fitzwilliam 138 kV	Rebuild line with higher-capacity conductor	\$9,437,000	100% (\$9,437,000)
Auburn Rd – Prides Corner 345 kV	Rebuild line with higher-capacity conductor	\$12,950,000	45% (\$5,798,000)
Westward – Auburn Road Substation 138 kV L	Rebuild line with higher-capacity conductor	\$6,823,000	33% (\$2,244,000)
Tuco – John Day Substation 138 kV L	Rebuild line with higher-capacity conductor	\$1,563,000	18% (\$277,000)
Matanzas – John Day 138 kV Line	Rebuild line with higher-capacity conductor	\$2,497,000	18% (\$443,000)
Auburn Road 500/345 kV TX	Replace transformer with higher-rated unit	\$34,175,000	44% (\$15,161,000)
Morgantown Energy Facility – Prides Corner 345 kV L	Rebuild line with higher-capacity conductor	\$12,412,000	59% (\$7,353,000)
Ronco 500-345 kV Auto	Replace transformer with higher-rated unit	\$27,436,000	41% (\$11,145,000)
Cavendish – Westward 138 kV	Rebuild line with higher-capacity conductor	\$4,424,000	33% (\$1,455,000)
	Total allocated to project		\$134,024,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.